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(*
                            CS 51 Lab 15
                          Native Lazy Streams
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(*.....................................................................
An implementation of lazy streams using OCaml's native `Lazy` module,
along with some useful functions.

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                     YOU SHOULD NOT EDIT THIS FILE.
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 *)

type 'a stream_internal = Cons of 'a * 'a stream
 and 'a stream = 'a stream_internal Lazy.t ;;

(* head strm -- Returns the first element of `strm`. *)
val head : 'a stream -> 'a ;;

(* tail strm -- Returns a stream containing the remaining elements of
   `strm`. *)
val tail : 'a stream -> 'a stream ;;

(* first n strm -- Returns a list containing the first `n` elements
   of the `strm`. *)
val first : int -> 'a stream -> 'a list ;;

(* smap fn strm -- Returns a stream that applies the `fn` to each
   element of `strm`. *)
val smap : ('a -> 'b) -> 'a stream -> 'b stream ;;

(* smap2 fn strm1 strm2 -- Returns a stream that applies the `fn` to
   corresponding elements of `strm1` and `strm2`. *)
val smap2 : ('a -> 'b -> 'c) -> 'a stream -> 'b stream -> 'c stream ;;

(* sfilter condition strm -- Returns a stream of all elements of
   `strm` for which `condition` holds. *)
val sfilter : ('a -> bool) -> 'a stream -> 'a stream ;;
```